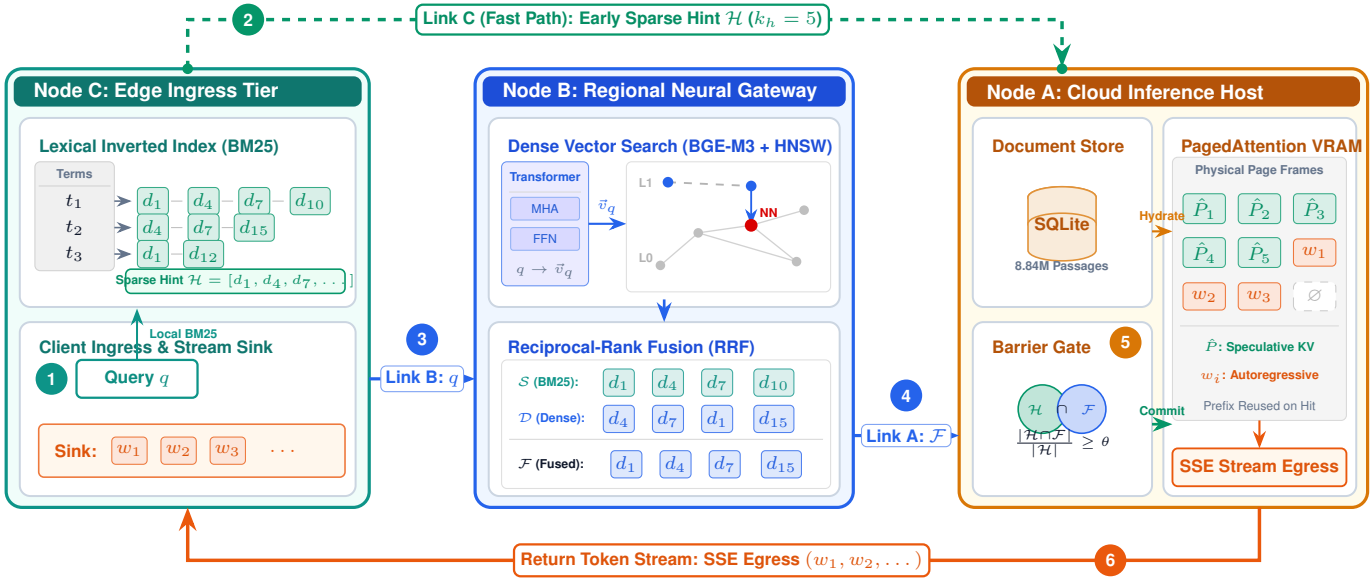

2
Link C (Fast Path): Early Sparse Hint $\mathcal{H}$ ($k_h = 5$)
Node C: Edge Ingress Tier
Lexical Inverted Index (BM25)
Terms
t_1
t_2
t_3
d_1 d_4 d_7 d_{10}
d_4 d_7 d_{15}
d_1 d_{12}
Sparse Hint $\mathcal{H} = [d_1, d_4, d_7, \ldots]$
Local BM25
Client Ingress & Stream Sink
1
Query q
Sink: w_1 w_2 w_3 ...
3
Link B: q
Node B: Regional Neural Gateway
Dense Vector Search (BGE-M3 + HNSW)
Transformer
MHA
FFN
$q \rightarrow \vec{v}_q$
$\vec{v}_q$
L1
L0
NN
Reciprocal-Rank Fusion (RRF)
$\mathcal{S}$ (BM25): d_1 d_4 d_7 d_{10}
$\mathcal{D}$ (Dense): d_4 d_7 d_1 d_{15}
$\mathcal{F}$ (Fused): d_1 d_4 d_7 d_{15}
4
Link A: $\mathcal{F}$
Node A: Cloud Inference Host
Document Store
SQLite
8.84M Passages
Hydrate
PagedAttention VRAM
Physical Page Frames
$\hat{P}_1$ $\hat{P}_2$ $\hat{P}_3$
$\hat{P}_4$ $\hat{P}_5$ w_1
w_2 w_3 $\varnothing$
$\hat{P}$: Speculative KV
w_i: Autoregressive
Prefix Reused on Hit
Barrier Gate
5
$\frac{|\mathcal{H} \cap \mathcal{F}|}{|\mathcal{H}|} \geq \theta$
Commit
SSE Stream Egress
Return Token Stream: SSE Egress $(w_1, w_2, \ldots)$
6